THE IMAGE OF LOWER SHRIEK INTO PUSHFORWARD FOR A SEPARATED ETALE MORPHISM IS THE PROPER-SUPPORT SUBSHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: two asserted steps sit on displayed constructions with named anchors: the presheaf image is 'clearly contained' in the proper-support subsheaf (proof lines 436-437, containment of one displayed construction in the subsheaf defined in lemma-f-shriek-separated), and the formal sum constructed in the decomposition 'maps to the restriction of s by construction' (proof lines 457-461, directly checkable from the two displayed constructions). The author's remark after the statement (statement lines 426-428, an alternative proof strategy via stalk descriptions) is an editorial aside, not an omission. No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Lower shriek into pushforward for separated etale morphisms (Etale Cohomology, Lemma etale-cohomology-lemma-shriek-into-star-separated-etale, canonical tag 0F4L). cited at statement lines 419-423 and proof lines 431-437, 458-460. Supplies the injective map j_!F -> j_*F, its presheaf-level construction, and the formula for j_{p!}F used in the converse.
- Proper-support subsheaf (Lemma lemma-f-shriek-separated, canonical tag 0F4X). cited at statement line 423 and proof lines 436-437. The subsheaf identified with the image.
- Etale-local decomposition (Etale Morphisms, Lemma etale-lemma-etale-etale-local-technical, canonical tag 04HM). cited at proof lines 447-453. Provides the etale-local splitting of U into copies of the base over which the support lies, enabling the pullback construction.

Source reference. Original lemma 0F4L: sources/stacks-chow/tags.txt, source lines 18906–18906. Cited at statement lines 419-423 and proof lines 431-437, 458-460: supplies the injective map, its presheaf construction, and the formula for j_{p!}F.

Source reference. Original lemma 04HM: sources/stacks-chow/tags.txt, source lines 5873–5873. Cited at proof lines 447-453: the etale-local splitting U = W disjoint union pieces isomorphic to the base.

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Source reference. Original lemma 0F4X: sources/stacks-chow/tags.txt, source lines 18917–18917. Cited at statement line 423 and proof lines 436-437: the subsheaf identified with the image of j_!F -> j_*F.

Source reference. Original lemma 02LS: sources/stacks-chow/tags.txt, source lines 3419–3419. Cited at proof lines 441-444: the support Z proper over V is finite with closed image Z'.

Lemma 0.2. *Let $j : U \to X$ be a separated étale morphism. Let $\mathcal{F}$ be an abelian sheaf on $U_{étale}$. The image of the injective map $j_!\mathcal{F} \to j_*\mathcal{F}$ of Étale Cohomology, Lemma 0F4L is the subsheaf of Lemma 0F4X.*

Proof. The construction of $j_!\mathcal{F} \to j_*\mathcal{F}$ in the proof of Étale Cohomology, Lemma 0F4L is via the construction of a map $j_{p!}\mathcal{F} \to j_*\mathcal{F}$ of presheaves whose image is clearly contained in the subsheaf of Lemma 0F4X. Hence since $j_!\mathcal{F}$ is the sheafification of $j_{p!}\mathcal{F}$ we conclude the image of $j_!\mathcal{F} \to j_*\mathcal{F}$ is contained in this subsheaf. Conversely, let $s \in j_*\mathcal{F}(V)$ have support Z proper over V. Then $Z \to V$ is finite with closed image $Z' \subset V$, see More on Morphisms, Lemma 02LS. The restriction of s to $V \setminus Z'$ is zero and the zero section is contained in the image of $j_!\mathcal{F} \to j_*\mathcal{F}$. On the other hand, if $v \in Z'$, then we can find an étale neighbourhood $(V', v') \to (V, v)$ such that we have a decomposition $U_{V'} = W \amalg U'_1 \amalg \ldots \amalg U'_n$ into open and closed subschemes with $U'_i \to V'$ an isomorphism and with $T_{V'} \subset U'_1 \amalg \ldots \amalg U'_n$, see Étale Morphisms, Lemma 04HM. Inverting the isomorphisms $U'_i \to V'$ we obtain n morphisms $\varphi'_i : V' \to U$ and sections s'_i over V' by pulling back s. Then the section $\sum(\varphi'_i, s'_i)$ of $j_{p!}\mathcal{F}$ over V', see formula for $j_{p!}\mathcal{F}(V')$ in proof of Étale Cohomology, Lemma 0F4L, maps to the restriction of s to V' by construction. We conclude that s is étale locally in the image of $j_!\mathcal{F} \to j_*\mathcal{F}$ and the proof is complete. □